E-COMMERCE PROFIT LEAKAGE ANALYSIS

Where the marketplace leaks profit

100,000 orders, three leaks, one recovery plan

Late deliveries | Freight cost | Weak sellers

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EXECUTIVE SUMMARY

Three leaks, one clear priority

6.8%

of orders arrive late

Reviews fall from about 4.3 to about 1.7 when delivery is 8+ days late

$272K

of freight above the average rate

Freight costs $2.25M in total, 16.6% of revenue

$751K

of revenue tied to 57 weak sellers

That is 5.5% of revenue, and 23% comes from sellers with under 30 orders

Priority: fix freight first. It is the largest leak we can put a number on.

THE BUSINESS QUESTION

Where does an online marketplace lose revenue, and how much can it recover?

01

Delivery

Do late orders cost ratings and repeat sales?

02

Freight

Where does shipping cost exceed the marketplace average?

03

Sellers

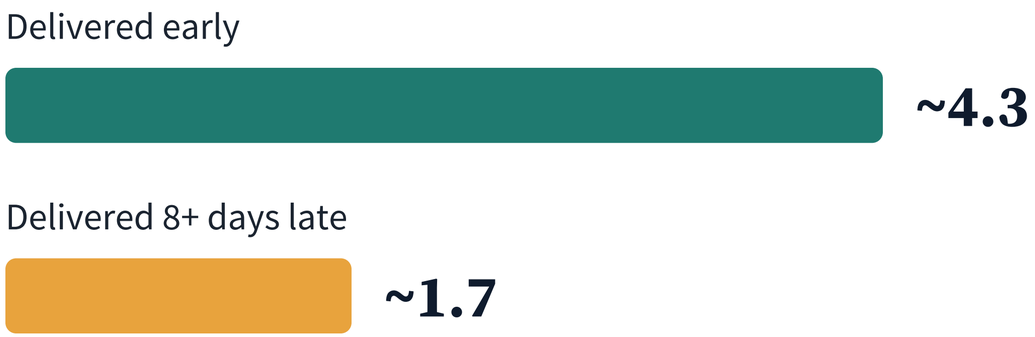
Which sellers put revenue at risk?

Scope: 100,000 orders · Olist Brazilian E-Commerce Public Dataset (Kaggle) · currency in $

LEAK 1 · DELIVERY

Late orders hurt ratings, not retention

Average review score by delivery timing (scale of 1 to 5)



6.8%

of orders arrive late

$9.7K

repeat-purchase loss, only 0.07% of revenue

Most customers buy once, so lateness shows up in reviews rather than in repeat sales.

LEAK 2 · FREIGHT

Freight is the largest quantifiable leak

$272K

of freight sits above the marketplace average rate

Total freight is $2.25M, which is 16.6% of revenue

Concentrated in three categories

furniture_decor | housewares | bed_bath_table

26–28%

of revenue goes to freight in remote states RR and MA

LEAK 3 · SELLERS

Seller risk sits with a small group

Share of total revenue

Revenue from the 57 weak sellers **5.5%**

Revenue from sellers with under 30 orders **23%**

57

weak sellers flagged

$751K

revenue they account for

The risk is concentrated, so action can target a short list of sellers.

PRIORITISATION

Where to act first

Leak	What the data shows	Size	Priority
Freight	Above-average rates, led by three product categories	$272K above average	**1 · Fix first**
Sellers	57 weak sellers; 23% of revenue from sellers under 30 orders	$751K revenue at risk	**2 · Review list**
Delivery	6.8% of orders late; reviews fall to ~1.7 at 8+ days	$9.7K repeat-purchase loss	**3 · Monitor**

Freight is cost above the marketplace average; the seller figure is revenue exposure. The sizes are not additive. Ranking is a proposed order based on size.

RECOMMENDATIONS · PROPOSED

Three moves to recover value

1

Re-price freight in the three heaviest categories

Start with furniture_decor, housewares and bed_bath_table

Target: $272K

2

Put the 57 weak sellers on a watchlist

Review thin-history sellers, who bring in 23% of revenue

At stake: $751K

3

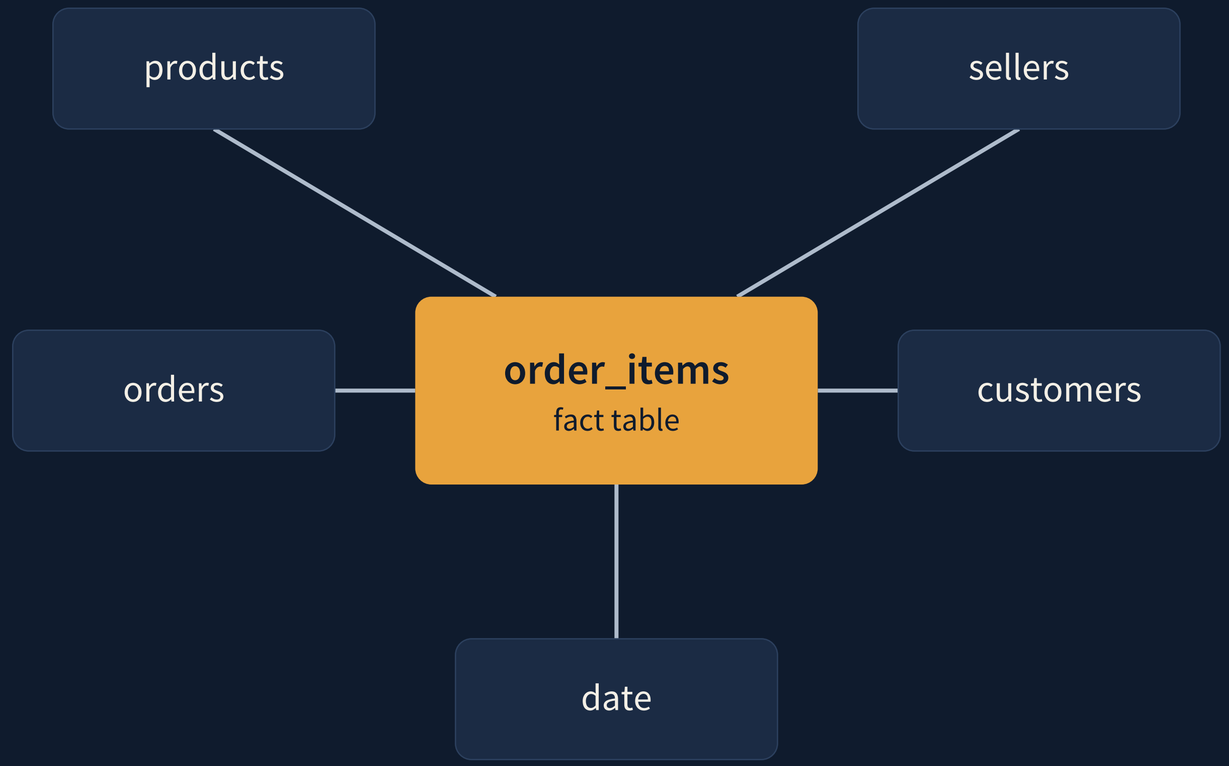
Flag orders at risk of arriving 8+ days late

Protect ratings, which fall to about 1.7 at that delay

Affects: 6.8% of orders

METHOD

How the dashboard was built



Clean

Power Query fixes types, merges tables, and keeps reviews and payments to one row per order

Model

A star schema with an order_items fact table and a date dimension

Measure

30+ DAX measures, calculated columns for delay bucket and seller tier, and a what-if parameter

THE TAKEAWAY

Start with freight, then watch sellers and delivery

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github.com/Singhtanusree/ecommerce-profit-leakage-powerbi